

Assignment 1

Steps to Set Up and Run a Google App Engine (GAE) Application Using Python 2.7

1. Install Required Software

- Install **Google App Engine SDK**.
- Install **Python 2.7**.

2. Configure Preferences

- Open **Google App Engine Launcher**.
- Go to **Edit > Preferences**.
- Set the following paths:
 - **Python Path:** C:\Python27\pythonw.exe
 - **App Engine SDK Path:** C:\Program Files (x86)\Google\google_appengine
- Click **OK** to save the preferences.

3. Create a New Application

- Go to your **Desktop** and create a **new folder** (this will be your app workspace).
- Open **Google App Engine Launcher**, click on **File > Create New Application**.
- Select the folder you just created on the Desktop.
- Give your application a **name**.
- Click **Create**.

4. Run the Application

- In the Google App Engine Launcher, the application will now be listed.
- Click on the **application name** and then click **Run**.

5. Edit the Application Code

- To modify the default (inbuilt) code:
 - Open the **folder** you created on the Desktop.
 - Inside, you'll find another folder
 - Open that folder in **Visual Studio Code**.
 - Locate and open the main.py file.
 - Edit the code as desired.

Assignment 2

Steps to Set Up and Run a Google App Engine (GAE) Application Using Python 2.7

1. Install Required Software

- Install **Google App Engine SDK**.
- Install **Python 2.7**.

2. Configure Preferences

- Open **Google App Engine Launcher**.
- Go to **Edit > Preferences**.
- Set the following paths:
 - **Python Path:** C:\Python27\pythonw.exe
 - **App Engine SDK Path:** C:\Program Files (x86)\Google\google_appengine
- Click **OK** to save the preferences.

3. Create a New Application

- Go to your **Desktop** and create a **new folder** (this will be your app workspace).
- Open **Google App Engine Launcher**, click on **File > Create New Application**.
- Select the folder you just created on the Desktop.
- Give your application a **name**.
- Click **Create**.

4. Run the Application

- In the Google App Engine Launcher, the application will now be listed.
- Click on the **application name** and then click **Run**.

5. Edit the Application Code

- To modify the default (inbuilt) code:
 - Open the **folder** you created on the Desktop.
 - Inside, you'll find another folder (the actual app directory).
 - Open that folder in **Visual Studio Code**.
 - Add another folder as templates in it and in templates add index.html file and in it add below to-do list code in it :

```

<!DOCTYPE html>
<html>
<head>
  <title>Simple To-Do List</title>
</head>
<body>

  <h2>My To-Do List</h2>

  <input type="text" id="task" placeholder="Enter a task">
  <button onclick="addTask()">Add</button>

  <ul id="list"></ul>

  <script>
    function addTask() {
      var task = document.getElementById("task").value;
      if (task !== "") {
        var li = document.createElement("li");
        li.textContent = task;
        document.getElementById("list").appendChild(li);
        document.getElementById("task").value = "";
      }
    }
  </script>

</body>
</html>

```

- And change the main.py code to below code :

```

import webapp2

class MainHandler(webapp2.RequestHandler):
    def get(self):
        self.response.headers['Content-Type'] = 'text/html'
        with open('templates/index.html', 'r') as f:
            self.response.write(f.read())

app = webapp2.WSGIApplication([
    ('/', MainHandler)
], debug=True)

```

Assignment 3

CloudSim Setup Instructions

1. Java Installation

1.1. Check if Java is already installed on your system:

Open Command Prompt and run:

```
java -version
```

1.2. If Java is not installed, download it from the official website:

<https://www.java.com/en/download/manual.jsp>

1.3. Install the Java setup by following the installation instructions.

1.4. Set the Environment Variable for Java:

- Right-click on *This PC* or *My Computer* → Click on *Properties*
- Click *Advanced system settings* → Click on *Environment Variables*
- Under *System Variables*, find and edit the `Path` variable
- Add the path to the Java `bin` folder (e.g., `C:\Program Files\Java\jdk1.8.0_xx\bin`)
- Click OK and close all windows

2. Download CloudSim and Additional JAR Files

2.1. Download CloudSim 3.0.3 from:

<https://code.google.com/archive/p/cloudsim/downloads>

2.2. Download Commons Math 3 JAR file from:

https://commons.apache.org/math/download_math.cgi

2.3. Extract both downloaded files (CloudSim and Commons Math JAR)

2.4. Copy the `commons-math3-3.6.1.jar` (or the version you downloaded)

Paste it into the `jars` folder inside the CloudSim directory

3. Eclipse IDE Installation

3.1. Download Eclipse IDE from:

<https://www.eclipse.org/downloads/packages/>

3.2. If you have 64-bit Java installed, download the 64-bit version of Eclipse.

If you have 32-bit Java installed, download the 32-bit version of Eclipse.

3.3. Install Eclipse by running the setup and completing the installation process.

4. Run CloudSim in Eclipse

4.1. Open Eclipse IDE

4.2. Create a new Java Project:

- Go to *File* → *New* → *Java Project*
- Enter the project name (e.g., `CloudSimProject`)
- Click *Finish*

4.3. Copy the contents of the CloudSim `source` folder

Paste them into the `src` folder of the Eclipse project

4.4. Add External JARs:

- Right-click the project → *Build Path* → *Configure Build Path*
- Go to the *Libraries* tab → Click *Add External JARs*
- Add:
 - `commons-math3-3.6.1.jar`
 - All JAR files from the CloudSim `jars` folder

4.5. Click *Apply and Close*

4.6. Right-click on a main file (like `CloudSimExample1.java`)

Click *Run As* → *Java Application*

Assignment 5

Ubuntu Installation using Oracle VirtualBox

1. Install Oracle VirtualBox

1.1. Download VirtualBox from the official website:

<https://www.virtualbox.org/wiki/Downloads>

1.2. Install VirtualBox by running the downloaded setup file and following the installation instructions.

2. Download Ubuntu ISO Image

2.1. Download the latest Ubuntu Desktop ISO file from the official website:

<https://ubuntu.com/download/desktop/thank-you?version=24.04.2&architecture=amd64&ts=true>

3. Open VirtualBox

3.1. Launch the VirtualBox application from your system.

4. Create a New Virtual Machine

4.1. Click the **New** button in VirtualBox.

4.2. Follow the prompts:

- Enter a **name** for your Ubuntu virtual machine.
- Select the **Ubuntu ISO image** you downloaded earlier.
- Check the box labeled **Skip unattended installation**.
- Click **Finish** to complete the setup.

5. Start Ubuntu

5.1. On the left-hand side of the VirtualBox window, locate the newly created virtual machine with the name you provided.

5.2. Click to open and start the Ubuntu virtual machine.

6. Ubuntu is Now Installed in VirtualBox

You can now begin using Ubuntu inside the VirtualBox environment.

Assignment 4

Procedure to Transfer Files Between Virtual Machines

1. Using Shared Folders in VirtualBox

Steps:

1. Set up a Shared Folder:

- Open **VirtualBox** and select the VM to which you want to add a shared folder.
- Click on **Settings** → **Shared Folders**.
- Click on the + icon on the right side.
- Select a folder from your host system that you want to share and set permissions (e.g., *Auto-mount*, *Read-only*).
- Click **OK** to save the shared folder configuration.

2. Access the Shared Folder on the Guest VM:

- Start the VM and log into Ubuntu or the OS installed in the VM.
- Open the **Terminal** and run:

```
bash
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sudo mount -t vboxsf <shared_folder_name> /mnt
```

- This will mount the shared folder to `/mnt`. You can now access and transfer files from this folder.

3. Transfer Files:

- You can copy files from the shared folder to any location inside the VM or vice versa.

2. Using SCP (Secure Copy Protocol) for Linux VMs

Steps:

1. Install SSH Server:

- Ensure the SSH server is installed on both VMs by running the following command on both VMs:

```
bash
CopyEdit
sudo apt update
sudo apt install openssh-server
```

- Confirm that SSH is running by checking the status:

```
bash
CopyEdit
sudo systemctl status ssh
```

2. Find the IP Address of the Source VM:

- On the source VM, find the IP address by running:

```
bash
CopyEdit
ip addr show
```

- Look for the IP address (usually in the format 192.168.x.x or 10.x.x.x).

3. Transfer Files Using SCP:

- On the destination VM, use the following command to copy files:

```
bash
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scp username@source_vm_ip:/path/to/file /path/to/destination
```

- Replace `username` with the username of the source VM, `source_vm_ip` with the IP address of the source VM, and specify the source and destination file paths.
- Example:

```
bash
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scp user@192.168.1.100:/home/user/file.txt
/home/destination_user/
```

4. Enter Password:

- You will be prompted to enter the password for the source VM. After entering the password, the file will be transferred.

3. Using FTP (File Transfer Protocol)

Steps:

1. Install an FTP Server on the Source VM:

- You can install `vsftpd` (Very Secure FTP Daemon) on the source VM by running:

```
bash
CopyEdit
sudo apt update
sudo apt install vsftpd
```

- After installation, start and enable the FTP service:

```
bash
CopyEdit
sudo systemctl start vsftpd
sudo systemctl enable vsftpd
```

2. Configure the FTP Server:

- Edit the `vsftpd` configuration file:


```
bash
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sudo nano /etc/vsftpd.conf
```

- Make sure the following options are set:
 - anonymous_enable=NO
 - local_enable=YES
 - write_enable=YES

3. Restart FTP Service:

```
bash
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sudo systemctl restart vsftpd
```

4. Find the IP Address of the Source VM:

- Run `ip addr show` on the source VM to get its IP address.

5. Transfer Files Using an FTP Client:

- On the destination VM, use an FTP client like `ftp` or `FileZilla` to connect to the source VM using its IP address.
- Example command:

```
bash
CopyEdit
ftp source_vm_ip
```

- After logging in with the credentials, you can use FTP commands to upload/download files.

4. Using NFS (Network File System)

Steps:

1. Install NFS Server on Source VM:

```
bash
CopyEdit
sudo apt update
sudo apt install nfs-kernel-server
```

2. Create a Directory to Share:

```
bash
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sudo mkdir /shared_folder
sudo chmod 777 /shared_folder
```

3. Configure NFS Exports:

- Edit the NFS exports file:

```
bash
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sudo nano /etc/exports
```

- Add the following line to share the folder:

```
bash
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/shared_folder *(rw, sync, no_subtree_check)
```

4. Export the Shared Directory:

```
bash
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sudo exportfs -a
```

5. Start the NFS Server:

```
bash
CopyEdit
sudo systemctl restart nfs-kernel-server
```

6. Mount the Shared Directory on the Destination VM:

- On the destination VM, run:

```
bash
CopyEdit
sudo mount source_vm_ip:/shared_folder /mnt
```

7. Transfer Files:

- You can now copy files to/from the shared folder.

5. Using USB Pass-Through (for VirtualBox)

Steps:

1. **Connect the USB Drive to the Host System:**
 - Plug in the USB drive to your host machine.
2. **Enable USB Passthrough:**
 - In VirtualBox, go to the settings of the VM.
 - Under **USB**, enable USB controller and add the USB device.
3. **Access the USB Drive in the Guest VM:**
 - Once the USB is passed through, you can access it from the guest VM like any regular USB device.
4. **Transfer Files:**
 - Copy the files from the USB drive to the destination VM or vice versa.